

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: Department of Electrical Engineering

2. Subject Name: Algorithms, AI and ML Laboratory

L-T-P: 1-0-3

Credits: 3

3. Pre-requisites: None

4. Syllabus and reference books:

Syllabus:

Divide and Conquer. Sorting and Order Statistics. Search. Graph and Tree Algorithms. Graph and Flows. Greedy Algorithms. Roots of Nonlinear Equations. Solution of Linear Equations. Eigenvalue Solvers and Applications. Decision Trees and Random Forest. Neural Networks and Deep Learning. Recurrent Neural Networks.

Reference Books:

- 1) T. H. Cormen, C. E. Leiserson, R. L. Rivest, C. Stein, Introduction to Algorithms, PHI, 3/e, 2009.
- 2) S. C. Chapra, R. P. Canale, Numerical methods for engineers, McGraw Hill, 7/e, 2015.
- 3) T. M. Mitchell, Machine Learning, McGraw Hill, 1/e, 1997.
- 4) S. Haykin, Neural Networks and Learning Machines, Pearson, 3/e, 2009.
- 5) H. Kinsley, D. Kukiela, Neural Networks from Scratch in Python, NNFS.io, 2023.

5. Lecture-wise break-up:

Sl. No.	Topic	No. of Lectures	No. of Lab Hours
1.	Divide and Conquer <i>Maximum-subarray problem. Strassen's algorithm for matrix multiplication. Asymptotic notation and complexity analysis.</i>	1	3
2.	Sorting and Order Statistics <i>Insertion sort. Heapsort. Quicksort. Counting sort. Radix sort. Bucket sort. Medians and order statistics.</i>	1	3

3.	Search <i>Stacks, queues, linked list, representing rooted trees. Binary search tree.</i>	1	3
4.	Graph and Tree Algorithms <i>Graph representation. Breadth-first search. Depth-first search. Minimum spanning tree. Kruskal and Prim algorithm.</i>	1	3
5.	Graph and Flows <i>Bellman-Ford algorithm. Dijkstra's algorithm. Shortest path and matrix multiplication with Floyd-Warshall algorithm. Ford-Fulkerson method.</i>	1	3
6.	Greedy Algorithms <i>Huffman codes. Matroids and greedy search. Task scheduling.</i>	1	3
7.	Roots of Nonlinear Equations <i>Bisection method. False position method. Newton-Raphson method. Secant method.</i>	1	3
8.	Solution of Linear Equations <i>Gauss elimination method, and with pivoting. Triangular factorization method. Matrix inversion method. Gauss-Seidel iterative method.</i>	1	3
9.	Eigenvalue Solvers and Applications <i>Eigenvalue problems. Fadeev-Leverrier method. Power method. Singular value decomposition (SVD). Principal component analysis (PCA).</i>	1	3
10.	Decision Trees and Random Forest <i>Binary decision trees. Random forest. Bootstrap aggregation. Tree pruning. Tree traversal and representation.</i>	1	3
11.	Neural Networks and Deep Learning <i>Feedforward neural networks. Convolution neurons. Activation functions. Loss functions. Training of neural networks. Autoencoders and PCA. Handwritten digit and image classification.</i>	1	3
12.	Recurrent Neural Networks <i>Recurrent neural networks. Long short-term memory (LSTM). Bi-RNN and Bi-LSTM. Application to electrical power demand estimation.</i>	1	3
Total number of hours		12	36